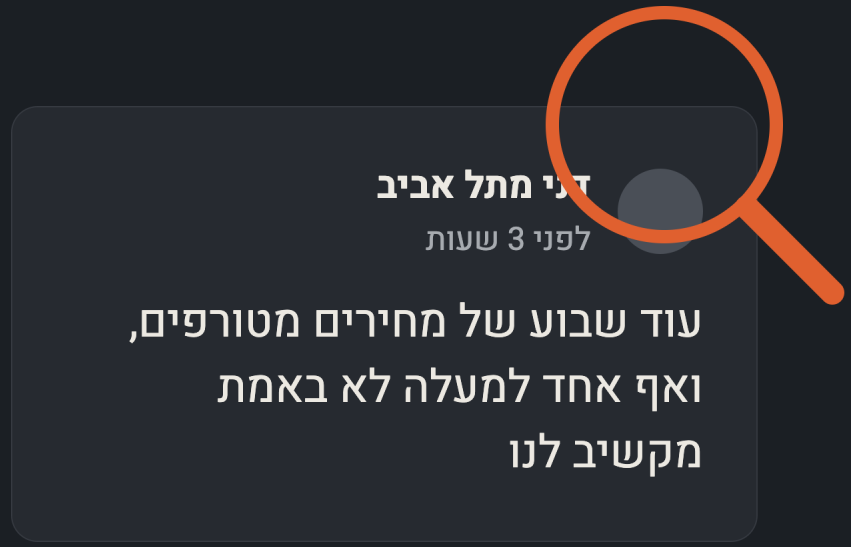
Hebrew Impostor Detection

Spotting LLM-written posts that pose as Israelis on Hebrew social media

Ido Brenner · Oren Yehezkel

Project proposal · Large Language Models and Their Applications · Bar-Ilan University



Impostor or authentic?

Motivating use case

Foreign influence operations run profiles that pose as Israelis.

More and more of what they post is written by LLMs.

Why it matters

Fake “Israeli voices” can steer public discussion on Hebrew social media.

Why it’s hard

LLM text is fluent and on-topic. There is no labeled Hebrew data for this task, and a classifier can learn topic or model style instead of impersonation.

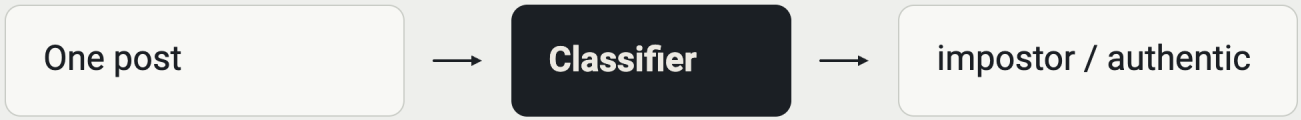
How it’s handled today

Platform moderation and manual investigation. Generic AI-text detectors answer “AI or human?”, not “is this really an Israeli?”.

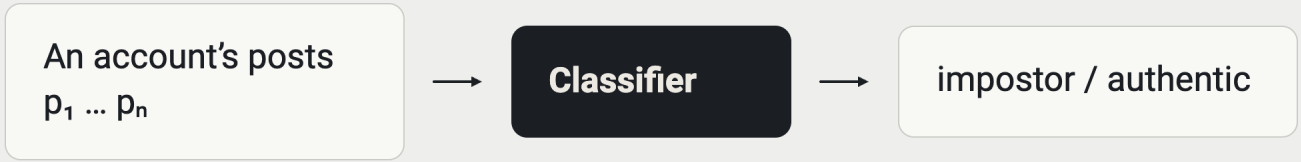
Task description

Binary classification: given *x*, predict *y* ∈ {impostor, authentic}

Post level



Account level



Account definition (n, how synthetic accounts are composed): **TBD**

What's new

Hebrew

Hebrew social-media posts and Hebrew models

Impersonation, not just "AI text"

LLM content written as if by an Israeli

Cross-generator generalization

Does a detector trained on one LLM's posts catch another LLM's?

Models & methods

Generate & collect
LLMs write posts from Israeli personas; real posts collected

→

Match topics
Same topics in both classes, against shortcut learning

→

Train & prompt
Four methods on the same data

→

Evaluate
Post and account level, across generators

Trained on our data

TF-IDF + logistic regression	Baseline
DictaBERT / AlephBERT	Fine-tuned Hebrew encoders
LoRA on a small LLM	Fine-tuned · model **TBD**

Off-the-shelf

LLM zero-shot / few-shot

No training · model **TBD**

Data

Impostor class • synthetic

What

Posts written by LLMs prompted with Israeli personas

Generators

Several LLMs. Which ones, and which are held out: **TBD**

Optional extension

Machine-translated Hebrew from Arabic, Persian or English posts. In v1: **TBD**

Authentic class • real

What

Real posts by Israelis

Source

An existing dataset if one exists, else manual collection by Ido and Oren: **TBD**

Recorded per post

Source and topic

Labels

Known by construction: no manual class labeling

Topic matching

Same topics in both classes, so topic can't give the answer away

Metrics & KPIs

Classification

F1 and AUC, at post level and at account level

Cross-generator

Train on one generator, test on each other one, plus leave-one-generator-out

Synthetic data quality

Checked at each stage. Specific measures: **TBD**

Ground truth

Known by construction: how a post was produced sets its label

Train ↓ · Test →	Gen A	Gen B	Gen C
Gen A	in-dist.	cross	cross
Gen B	cross	in-dist.	cross
Gen C	cross	cross	in-dist.
All others	held out	held out	held out

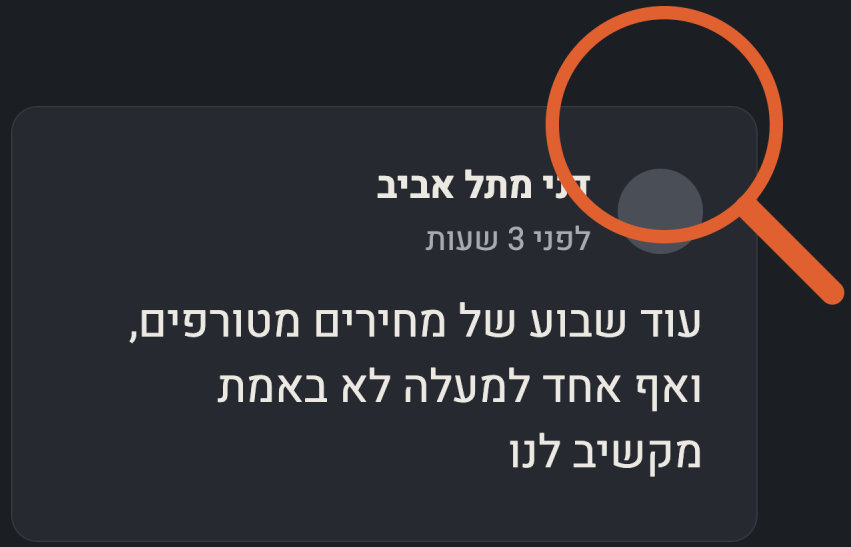
Each cell: F1 / AUC. Last row: leave-one-generator-out. Generators A–C are placeholders: models and count **TBD**

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